

PAY-PER-USE CHARGING INFRASTRUCTURE

44 kW grid connection. Three e-trucks. No problem.

How freight carriers and logistics operators electrify their fleet without tying up capital: fast charging plus battery storage with the PowerCube — billed per kilowatt-hour charged.

POWERCUBE * PAY-PER-USE * E-TRUCKS

THE MARKET ENVIRONMENT

“Companies now invest only to cover wear and tear”

A look at the books of more than a million small and medium-sized enterprises shows: revenues are stagnating, costs are rising – most firms now invest only to maintain existing assets, hardly at all in new capacity.

-0.23

NET FIXED-INVESTMENT RATIO
2025

Lowest level since 1990: for the first time since reunification, depreciation exceeds investment – the capital stock is shrinking.

+56 %

PLANNED BUSINESS CLOSURES

Increase in 2025 versus the previous year – driven by high costs, weak demand, bureaucracy and a lack of succession solutions.

≈ 0 %

POTENTIAL GROWTH

Combined with demographic change, Germany is heading towards zero potential growth by the end of the decade.

Asset-intensive sectors such as transport & logistics are hit twice by the investment backlog: fleets and infrastructure are ageing while the drivetrain transition demands speed.

“The electric truck is no longer a technology problem”

Prof. Dr. Markus Lienkamp (TU Munich) in an electrive interview: grid connections, depot strategies and costs will decide the ramp-up — not vehicle technology. Four key statements on depot charging.

DEPOT CHARGING

Backbone of the ramp-up

Three quarters of all routes are shorter than 219 km — depot charging alone can already cover 85% of energy demand.

GRID CONNECTION

The longest path

Connection upgrades can have lead times of several years — they cannot be made up later, only started early.

LOAD PEAKS

Storage as optimizer

Uncontrolled fast charging creates load peaks of several megawatts — with load management and battery storage, the same connection delivers considerably more.

CAPITAL

The investment hurdle

E-trucks win on operating costs — yet many projects currently stall at the initial investment in depot infrastructure.

“Waiting costs more than it saves” — the most attractive cost window is already open: the toll exemption for e-trucks only applies until mid-2031.

THE CHALLENGES

What slows down freight carriers and logistics operators today

The drivetrain transition is a done deal — yet on the way to an electric fleet, almost all companies face the same four hurdles.

REGULATION

CO₂ toll & cost pressure

CO₂-differentiated tolls and fleet emission limits make diesel more expensive year after year; shippers increasingly demand zero-emission transport.

INFRASTRUCTURE

Grid connection as bottleneck

Existing connections at logistics yards are not designed for fast-charging e-trucks — grid expansion is expensive and often takes years.

CAPITAL

Six-figure investments

Conventional charging infrastructure with transformer and grid expansion quickly reaches the mid six-figure range — on already thin margins.

RESOURCES

Know-how & operations

System design, grid operator coordination, DGUV inspections, monitoring: specialist tasks for which day-to-day haulage operations lack time and staff.

The result: many companies postpone electrification — losing time in the competition for tenders and skilled workers.

FROM THE FIELD

A real reference project shows the hurdles

A mid-sized freight carrier — around 70 employees, 42 tractor-trailers — plans to start with three e-trucks. The site's figures are typical for the entire industry.

44 kW

EXISTING GRID CONNECTION

Nowhere near sufficient for three fast-charging e-trucks.

~90.000 €

NEW GRID SUPPLY LINE

The grid operator quotes around €90,000 net for a new 250 m supply line.

€450 – 470k

CONVENTIONAL QUOTES

Two quotes received for conventional charging infrastructure with transformer.

Calculated conventionally: over half a million euros before the first e-truck charges — for many companies the end of electrification.

THE WAY OUT

Investing without investing

Logistics operators and freight carriers are under double pressure: margins in the core business are shrinking while the drivetrain transition demands large upfront outlays for e-trucks and charging infrastructure. This is exactly where pay-per-use comes in as a problem solver.

THE INVESTMENT DILEMMA

- ✗ **Six-figure CAPEX project** – conventional quotes for charging infrastructure in the reference project came to €450–470k.
- ✗ **Tied-up liquidity** – equity and credit lines are needed in the core business; firms now invest only “to cover wear and tear”.
- ✗ **Missing specialist know-how** – grid connection, storage technology and operation tie up resources that day-to-day haulage lacks.

PAY-PER-USE AS PROBLEM SOLVER

- ✓ **€0 investment** – the PowerCube arrives at your yard without CAPEX and without burdening the balance sheet.
- ✓ **Predictable costs** – billed per kilowatt-hour charged, cheaper than the energy costs of a diesel truck.
- ✓ **Modernization despite the investment backlog** – liquidity and credit lines stay in the core business; the fleet still becomes future-proof.

Electrification thus turns from an investment risk into a predictable operating expense – motivation to invest without straining your own balance sheet.

THE SOLUTION

The PowerCube solves the grid problem

Fast charging plus battery storage in one box: instead of an expensive new supply line, the integrated storage buffers the load peaks. The site charges e-trucks at high power – while drawing only a constant base load from the grid. Figures from the reference project:

80 kW**CONSTANT GRID
DRAW**

instead of costly grid
connection upgrades

750 kWh*BATTERY STORAGE**

Smart Charging Cube with
Powerco modules and
integrated HYC 400

*sized individually to the
charging profile

0 €**INVESTMENT**

Pay-per-use instead of
hardware purchase by the
customer

~16 wks**DELIVERY TIME**

from contract signing to
delivery

Reference example: a freight carrier with 70 employees and 42 tractor-trailers charges three e-trucks on the existing 44 kW connection – instead of €450–470k conventionally.

THE BUSINESS MODEL

Pay-per-use: pay per kilowatt-hour

The customer buys no hardware and bears no investment risk. AW Automotive provides the PowerCube; billing is purely consumption-based per kilowatt-hour charged. Terms from the reference project:

from 6 years	Term of the pay-per-use contract
from 20,000 kWh	Minimum charging volume / month
fixed kWh price	incl. maintenance, monitoring and DGUV inspection

Cheaper than diesel

The agreed kWh price plus charging electricity costs is below the energy costs per kilometre of a comparable diesel truck.

Cut charging costs further

AW Automotive additionally helps its customers reduce their own charging electricity costs – e.g. via an optimized electricity tariff and self-generated solar power, likewise on a pay-per-use basis.

360° perspective in the pay-per-use model

AW Automotive supports its customers from needs analysis to ongoing operation – with clear cost and planning certainty.

AW AUTOMOTIVE – WITH PAY-PER-USE

- ✓ **Needs & load analysis** – consumption, charging times and PV data as the basis of the system design.
- ✓ **Interface to the grid operator** – coordination to increase grid capacity.
- ✓ **Design as a charging solution** – CAPEX-free via pay-per-use, matched to the e-trucks' usage profile.
- ✓ **Management & monitoring** of grid connection works through to commissioning.
- ✓ **Contract & financing structure** – clear cost structure, 100% planning certainty.
- ✓ **Operating cost optimization** – cheaper electricity tariff & PV system, likewise via pay-per-use.
- ✓ **Ongoing support** with cost optimization and charging requirements.

MARKET STANDARD – WITHOUT PAY-PER-USE

- X **Quote on request** – no options for reducing grid expansion or peak load costs.
- X **Communication with the grid operator** usually remains with the customer.
- X **Standard concept** – maximum grid expansion instead of actual need; adjustments cost extra.
- X **External services** are not supported.
- X **Financing** must be arranged by the customer independently.
- X **Operating costs** – no advice; a separate business model requiring additional investment.
- X **No included resources** for cost optimization during the ongoing project.

The pay-per-use model in operation

In ongoing operation, AW Automotive remains the central point of contact – without pay-per-use, responsibility is spread across many shoulders.

AW AUTOMOTIVE – WITH PAY-PER-USE

- ✓ **One central point of contact** – for everything related to the PowerCube, the customer has a single port of call: AW Automotive.
- ✓ **DGUV V3 inspection** – required annually by law; included in the pay-per-use concept and carried out automatically.
- ✓ **End-to-end monitoring** – AW Automotive permanently monitors the PowerCube's function and condition online to react to faults without delay.
- ✓ **Third-party interfaces** – support for integrating fleet management software and depot-sharing models.
- ✓ **Regular operating data analyses** – identify further savings potential to reduce charging electricity costs.

MARKET STANDARD – WITHOUT PAY-PER-USE

- X **Many points of contact** – multiple manufacturers mean many anonymous hotlines; liability disputes between manufacturers are inevitable.
- X **DGUV V3 inspection** must be organized and scheduled by the customer, or commissioned via an additional service contract.
- X **Limited monitoring** – hardly feasible end to end with mixed manufacturers; the customer must act on faults themselves.
- X **Difficult third-party integration** – due to the multitude of combined manufacturers and technologies.
- X **Incomplete data** – incompatible user interfaces from different manufacturers hamper target/actual cost analyses.

THE ECOSYSTEM

A strong network behind every project

Pay-per-use works because vehicle, finance and energy partners interact — coordinated by AW Automotive as the central point of contact.

MAN Truck & Bus

VEHICLE PARTNER

Close cooperation on e-truck projects — from the vehicle to the right charging solution at the yard.

AW Automotive

TECHNOLOGY & OPERATION

Provider of the Smart Charging Cube — from design and production to ongoing operation.

Findustrial

FINANCING

Enables the pay-per-use model by combining intelligent software with refinancing partners.

Audi

STRATEGIC HUB

Contributes experience from the Audi charging hubs and supports the concept strategically.

PROVIDER & CONTACT

AW Automotive GmbH

Concepts for battery-storage-backed charging infrastructure



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